School of Software Engineering: MSc Intelligent Systems

Final Project Report

Multi-Agent Social Interaction AIS

Trajectory Prediction: Comparing Methods

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Abstract

Vessel trajectory prediction from AIS underpins automated collision avoidance, yet published results in this field cannot be compared with one another: each study reports on a private regional dataset, at its own prediction horizon, under its own preprocessing and metric conventions. We address this by evaluating five architectures under a single protocol — an ordinary least-squares baseline, TrAISformer, AIS-ACNet, GAT-LSTM, and Wayformer-AIS, a maritime adaptation of a multi-agent attention model — on the same corpus of operational AIS, at a common five-minute horizon sampled at 1 Hz, with identical splits and metric implementations and a held-out test set of 11,891 prediction windows.

The comparison produces a result the maritime literature does not currently surface: at this horizon three of the four learned architectures predict less accurately than constant-velocity extrapolation of the observed track. Only Wayformer-AIS improves on that baseline, reaching 15.57 m minADE_6 and 31.71 m minFDE_6 against 26.70 m and 51.43 m, and its margin widens as manoeuvre difficulty rises — from 3.2 m on stationary vessels to 49.5 m on sharp turns. It is also the only model in the comparison given the surrounding traffic as input rather than a single vessel's track. Stratified analysis locates the remaining error in underway vessels in dense traffic and in turns already in progress. The preprocessing pipeline, model configurations and evaluation scripts are released so that the comparison can be repeated and extended.

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1 Introduction

1.1 Motivation

The global maritime sector carries more than 80 % of world trade by volume, and vessel traffic continues to grow alongside expanding global supply chains. As ships become larger and operating margins tighter, there is mounting demand for autonomous and decision-support systems capable of reducing human error, improving situational awareness, and enforcing compliance with the International Regulations for Preventing Collisions at Sea (COLREGs) [International Maritime Organization, 1972].

At the core of any autonomous maritime system lies a classical autonomy pipeline: *perception* (detecting other vessels from radar, camera, or AIS), *prediction* (estimating where those vessels will be in the near future), and *planning* (computing a safe, COLREG-compliant manoeuvre for the ego vessel). Trajectory prediction is the bridge between perception and planning: a collision-avoidance planner cannot reason about a safe path without a probabilistic estimate of how nearby vessels will move [Kuwata et al., 2014, Vagale et al., 2021, Mahipala et al., 2025]. Constant-velocity assumptions, the default fallback in many deployed systems, break down precisely in the situations that matter most — congested ports, narrow channels, and overtaking or crossing encounters governed by COLREG rules. Accurate, interaction-aware trajectory prediction is therefore not an academic exercise but a safety-critical prerequisite for practical autonomous bridge deployment.

The Automatic Identification System (AIS) provides the natural data foundation for this prediction task [International Telecommunication Union, 2014]. Under IMO mandate, every large commercial vessel continuously broadcasts its MMSI, position, speed, and course at intervals as short as 2 seconds, producing a dense, global, and freely receivable record of maritime motion. The richness of this signal has driven a wave of data-driven trajectory prediction research over the past decade, spanning recurrent networks, graph models, and transformers (see Section 2).

This study is motivated in part by the operational context of Orca AI, a company that deploys AI-based situational awareness systems on commercial vessels worldwide. Orca AI's onboard AIS integration produces a large-scale, geographically diverse archive of real maritime encounters accumulated since 2019, offering an opportunity to train and evaluate prediction models on operational data of a kind not commonly available to academic research. The corpus used in this work is a subset of that archive. Crucially, the scenarios in this dataset are captured from the perspective of a navigating vessel (*ego vessel*) surrounded by up to tens of nearby *target vessels*, making the interaction structure — which vessel is on a collision course, which is merely passing — an integral part of the prediction

problem.

1.2 The Comparison Problem in Maritime AI

Despite a growing body of AIS prediction literature, a practitioner seeking to select or deploy a model faces a fundamental obstacle: the published results cannot be meaningfully compared. TrAISformer [Nguyen and Fablet, 2021] is evaluated on Finnish–Baltic AIS data with a 30-minute horizon and cross-entropy loss; AIS-ACNet [Shin et al., 2024] uses South Korean coastal data with 10-second prediction steps; GAT-LSTM [Zhao et al., 2023] reports next-step ADE on Chinese coastal waters; TPTrans [Wang et al., 2024] evaluates on a Chinese port dataset with its own normalisation scheme. Each paper applies its own preprocessing, defines its own evaluation horizon, and reports on a proprietary regional dataset that no other group can access.

The consequence is that the basic question — *does a transformer outperform a graph network for vessel trajectory prediction, and by how much?* — cannot be answered from the existing literature. Method comparisons within individual papers are invariably partial, comparing only against baselines the authors chose to re-implement, on their own data, under their own evaluation protocol.

This paper resolves the comparison problem by subjecting five architectures — from a naïve linear baseline to a multi-agent transformer — to an identical experimental protocol: the same proprietary AIS dataset, the same preprocessing pipeline, the same train/validation/test split, and the same evaluation metrics. The result is the first controlled, apples-to-apples comparison of maritime trajectory prediction methods on a large-scale, multi-year, globally diverse dataset.

1.3 Problem Statement

We consider a maritime scene in which several vessels manoeuvre simultaneously within mutual observation range, and ask how each of them will move over the following minutes. The input is the recent motion history of every vessel in the scene, reconstructed from AIS. The required output is not a single future track per vessel but a small set of plausible alternatives with associated probabilities, because a vessel approaching a waypoint or an encounter may legitimately hold its course, alter to port or starboard, or slow to a stop. Predictions are made independently for each vessel while conditioning on the observed histories of all its neighbours, so that interaction informs the forecast without requiring the model to commit to a single joint future for the whole scene. Accuracy is judged by how closely the best of the proposed alternatives matches what the vessel actually did.

Section 3 states this task formally, together with the input and output representations, the observation and prediction horizons, and the evaluation metrics used throughout the

paper.

1.4 Contributions

This paper makes the following contributions:

1. **AIS data pipeline:** A complete preprocessing toolkit for raw AIS CSV data, including deduplication, gap interpolation, coordinate projection, and UniTraj-compatible serialization.
2. **Unified maritime benchmark:** The first evaluation of multiple trajectory prediction architectures under identical conditions — shared train/validation/test splits, preprocessing, and metrics — on a large-scale proprietary AIS dataset.
3. **Method comparison:** Systematic evaluation of five approaches — a naïve OLS linear baseline, TrAISformer, AIS-ACNet, Wayformer-AIS, and GAT-LSTM [Zhao et al., 2023] — each adapted to and evaluated under identical conditions on the same maritime AIS dataset.
4. **Reproducibility:** All code, configuration, and evaluation scripts are released to enable future comparisons on the same or alternative AIS datasets.

2 Related Work

Table 1 lists the twelve works reviewed in this section. Its purpose is not to summarise their methods — each is discussed individually in the subsections that follow — but to make a single observation visible at a glance: *no two of these studies are evaluated under the same conditions*. Each maritime entry reports on a different regional dataset, at a different prediction horizon, under its own preprocessing and metric conventions, and several do not specify their evaluation setting in reproducible detail at all. The absence of any shared row in the final column is precisely the gap this paper addresses.

Table 1: Related work, with the evaluation setting each study reports. † = maritime / AIS domain; unmarked methods originate from road-vehicle or pedestrian prediction and are adapted or compared in this work. "Not reported" indicates the source does not specify the setting in reproducible detail. No two rows share an evaluation setting, which is why the published numbers cannot be compared directly.

Model / Reference	**Category**	**Reported evaluation setting**
Zhang et al. 2024 †	RNN	Coastal AIS traffic data; region and prediction horizon not reported.
AIS-ACNet [Shin et al., 2024] †	Dilated CNN	South Korean coastal AIS; 10 s prediction steps.
GAT-LSTM [Zhao et al., 2023] †	Graph	Chinese coastal AIS; next-step ADE only.
ST-ENAGCN [Zeng et al., 2024] †	Graph	AIS encounter dataset; region not reported.
TrAISformer [Nguyen and Fablet, 2021] †	Transformer	Finnish–Baltic AIS; 30 min horizon, cross-entropy loss.
TPTrans [Wang et al., 2024] †	Transformer	Chinese port AIS; custom normalisation scheme.
AIS-LLM [Park, 2025] †	LLM	Multi-task maritime evaluation; setting not reported.
Social GAN [Gupta et al., 2018]	Multi-Agent	Pedestrian trajectory benchmarks (non-maritime).
Wayformer [Nayakanti et al., 2022]	Multi-Agent	Waymo Open Motion Dataset (road vehicles).
UniTraj [Feng et al., 2024]	Framework	Road-vehicle benchmarks under unified splits and metrics.
Alam et al. 2025 †	Multi-Agent	Multi-vessel AIS scenes; setting not reported.
Mahipala et al. 2025 †	Rule-Aware	Restricted waterways with COLREG and planned-route priors.

2.1 Sequential and RNN-Based Methods

The earliest data-driven approaches to AIS trajectory prediction modelled the position stream of a single vessel as a discrete time series and applied recurrent architectures to learn its temporal dynamics. LSTM and GRU networks became the dominant baseline: Zhang et al. [2024] demonstrate that well-tuned LSTM and GRU encoders achieve competitive ADE and FDE on coastal AIS data, and establish the recurrent baseline that subsequent work aims to surpass. Shin et al. [2024] advance this line with AIS-ACNet, replacing the recurrent core with a *dual-encoder dilated causal convolutional network* (GWNet backbone) that processes position and kinematics (SOG, COG) in parallel

streams. Exponential dilation factors give AIS-ACNet a wide receptive field over the past trajectory while auxiliary regression heads on SOG and COG provide additional training signal.

A shared limitation of all methods in this family is the *single-vessel assumption*: each prediction is produced independently from the vessel's own past track, with no representation of nearby traffic. This is adequate in open-ocean settings but breaks down in the encounter situations — crossing, overtaking, head-on — that are precisely the scenarios of greatest safety relevance. A model blind to neighbouring vessels cannot anticipate that the ego ship will slow or alter course in response to them.

2.2 Graph-Based Interaction Models

To capture vessel-to-vessel influence, a second generation of methods applies graph neural networks (GNNs) over the set of vessels present in a scene. Zhao et al. [2023] propose GAT-LSTM, which constructs a directed graph whose nodes represent trajectory sub-segments and applies Graph Attention Network (GAT) layers as a spatial encoder before an LSTM temporal decoder. The attention mechanism allows the model to weight the influence of each neighbouring segment according to learned geometric relevance, outperforming pure LSTM baselines on next-step prediction tasks in Chinese coastal waters. Zeng et al. [2024] extend the graph paradigm to encounter-level reasoning with ST-ENAGCN, a spatio-temporal graph convolutional network that encodes both node (vessel) and edge (encounter pair) features using separate attention streams, enabling collision-risk-informed trajectory generation.

Despite their interaction-awareness, graph-based models share two structural weaknesses. First, they are *deterministic*: both GAT-LSTM and ST-ENAGCN produce a single future trajectory per vessel, providing no uncertainty quantification and no mechanism for the multi-modal outputs required by safety planners. Second, they are evaluated on narrow, region-specific datasets — a single port or coastal zone — making cross-study comparison impossible.

2.3 Transformer-Based Methods

Transformer architectures have become the method of choice for long-sequence modelling in natural language and, increasingly, in trajectory prediction. Nguyen and Fablet [2021] introduce TrAISformer, the first GPT-style autoregressive transformer trained end-to-end on raw AIS streams. A sparse, quantised representation of latitude, longitude, speed, and course is decoded token-by-token; the cross-entropy loss over discrete trajectory tokens provides a natural probabilistic interpretation. Wang et al. [2024] develop TPTrans, a position-aware transformer that models temporal correlations in the raw AIS feature

stream and reports improvements over LSTM baselines on a Chinese port dataset. Most recently, Park [2025] propose AIS-LLM, a large-language-model framework that unifies trajectory prediction, anomaly detection, and collision-risk assessment in a single pre-trained architecture, achieving strong results across multiple evaluation tasks.

The common thread across transformer-based maritime work is strong performance within a single private dataset paired with a near-total absence of cross-method comparisons. Each paper reports on data that no other group can access, uses a different preprocessing scheme, and often evaluates at a different temporal resolution. The basic question — does an autoregressive transformer outperform a dilated CNN at a 5-minute maritime prediction horizon? — remains unanswered in the literature.

2.4 Multi-Agent Prediction from Autonomous Driving

The autonomous-driving community has produced a mature ecosystem of multi-agent trajectory prediction models, evaluation benchmarks (Waymo Open Motion Dataset, nuScenes), and standardised metrics that the maritime field largely lacks. Gupta et al. [2018] introduce Social GAN, combining a graph-based social pooling mechanism with a generative adversarial loss to produce socially acceptable, multi-modal pedestrian trajectories — establishing the paradigm of interaction-aware probabilistic prediction. Nayakanti et al. [2022] advance this with Wayformer, a Perceiver-based bottleneck encoder-decoder that applies factorised attention over agents and time to achieve state-of-the-art performance on the Waymo benchmark with a compact, scalable design. The unified benchmark framework of Feng et al. [2024] further enables fair cross-model comparison on road-domain data. From the maritime side, Alam [2025] apply interaction-aware prediction to vessel collision-risk assessment, demonstrating that scene-level joint modelling improves both trajectory accuracy and risk quantification.

Road-domain models are not directly transferable to the maritime setting: they depend on lane graphs, road topology, and agent-type encodings that have no analogue at sea. Vessel dynamics are also qualitatively different — longer stopping distances, wind and current effects, slower turn rates — and maritime encounters are governed by COLREGs rather than traffic lights and yield signs. No prior work has systematically adapted and evaluated road-domain multi-agent architectures on a large-scale AIS dataset under controlled conditions.

2.5 Rule-Aware and COLREG-Informed Prediction

The International Regulations for Preventing Collisions at Sea (COLREGs) [International Maritime Organization, 1972] impose explicit right-of-way and manoeuvring obligations on vessels in encounter situations. A trajectory prediction model that is unaware of COL-

REGs may forecast physically plausible but regulation-violating futures, producing unsafe inputs to downstream collision avoidance planners [Kuwata et al., 2014, Vagale et al., 2021]. Several recent works attempt to bridge this gap. Mahipala et al. [2025] present a probabilistic prediction framework that integrates COLREG obligations, grounding hazards, and planned route information as explicit priors over the predictive distribution, achieving improved calibration in restricted waterways.

A limitation common to rule-aware approaches is their reliance on hand-engineered priors: encounter type must be classified and the applicable COLREG rule selected before prediction can proceed. These classifiers introduce their own errors and do not generalise gracefully to ambiguous or multi-vessel scenarios. Purely learned models, by contrast, can in principle discover COLREG-consistent patterns from data without explicit rule specification — as the anchoring deceleration example in Section 5.3 illustrates. Integrating learned kinematic context with explicit regulatory structure remains an open problem; this paper evaluates the learned-only end of the spectrum and leaves rule-augmented hybrid models as future work.

2.6 Benchmarking and Reproducibility

The road-domain trajectory prediction community has benefited enormously from shared benchmarks: the Waymo Open Motion Dataset and the nuScenes prediction challenge provide standardised splits, metrics, and leaderboards that make method comparisons meaningful. Feng et al. [2024] embody this ethos in UniTraj, a unified training and evaluation framework for road-vehicle trajectory models that enables fair cross-architecture comparison on identical data splits. No equivalent infrastructure exists for maritime AIS prediction: every published AIS model is evaluated on a private regional dataset under custom preprocessing and metric choices.

2.7 Positioning This Work

The six bodies of work reviewed above converge on a shared gap: despite a decade of AIS prediction research, no study has placed multiple architectures — spanning RNN, graph, transformer, and multi-agent paradigms — under a common experimental protocol on the same large-scale dataset. This paper fills that gap. We adapt five architectures (Baseline Linear OLS, TrAISformer, AIS-ACNet, Wayformer-AIS, and GAT-LSTM) to an identical 5-minute, 1 Hz maritime prediction task, train and evaluate them on the same proprietary Orca AI AIS archive with the same preprocessing pipeline, data split, and metrics, and provide all code and configuration for reproducibility. The result is the first controlled, apples-to-apples benchmark for AIS trajectory prediction.

3 Problem Formulation

3.1 Notation

Let $\mathcal{V} = \{1, \ldots, N\}$ denote the set of vessels present in a maritime encounter scene. Each vessel $i \in \mathcal{V}$ is observed over a past horizon of T_{past} timesteps. At each timestep t, the state of vessel i is described by a feature vector

$$\mathbf{x}_i^t = \big(x,\ y,\ v_x,\ v_y\big) \in \mathbb{R}^4,$$

where (x, y) is the position in meters relative to an ego-centric reference frame, and (v_x, v_y) are the corresponding Cartesian velocity components in m/s.

3.2 Task Definition

Given the past trajectory matrix $\mathbf{X}_i = [\mathbf{x}_i^1, \ldots, \mathbf{x}_i^{T_{\text{past}}}] \in \mathbb{R}^{T_{\text{past}} \times 4}$ for all vessels $i \in \mathcal{V}$, the prediction task is to learn a function

$$f_\theta\big(\mathbf{X}_1, \ldots, \mathbf{X}_N\big) \longrightarrow \big\{\hat{\mathbf{Y}}_i\big\}_{i=1}^{N},$$

where $\hat{\mathbf{Y}}_i$ is a probability distribution over future trajectories for vessel i covering T_{future} timesteps.

Following standard practice in multi-modal trajectory prediction [Nayakanti et al., 2022], the output is a Gaussian Mixture Model (GMM) with $K = 6$ modes. For each mode k, the model predicts a trajectory $\hat{\mathbf{y}}_i^{(k)} \in \mathbb{R}^{T_{\text{future}} \times 2}$ and an associated probability weight $\pi_i^{(k)}$, with $\sum_k \pi_i^{(k)} = 1$.

3.3 Encounter Definition

A maritime encounter is defined as a spatiotemporal cluster of vessels within a maximum range of 12 nautical miles (22 km) of one another over a sliding time window of $T_{\text{past}}+T_{\text{future}}$ seconds. Vessels outside this range are treated as non-interacting and excluded from the scene.

The default configuration used throughout this work is:

$$T_{\text{past}} = 300\,\text{s}\ (5\,\text{min}), \quad T_{\text{future}} = 300\,\text{s}\ (5\,\text{min}), \quad K = 6.$$

3.4 Evaluation Metrics

All models are evaluated using the following standard metrics [Feng et al., 2024]:

- **minADE$_K$** (minimum Average Displacement Error): the ℓ_2 distance between the best predicted mode and ground truth, averaged over all future timesteps:

$$\text{minADE}_K = \min_{k=1}^{K} \frac{1}{T_{\text{future}}} \sum_{t=1}^{T_{\text{future}}} \left\| \hat{\mathbf{y}}_t^{(k)} - \mathbf{y}_t \right\|_2$$

- **minFDE$_K$** (minimum Final Displacement Error): displacement error at the final predicted timestep only:

$$\text{minFDE}_K = \min_{k=1}^{K} \left\| \hat{\mathbf{y}}_{T_{\text{future}}}^{(k)} - \mathbf{y}_{T_{\text{future}}} \right\|_2$$

- **Miss Rate @ d m**: fraction of predictions where $\text{minFDE}_K > d$, reported at $d \in \{2, 10, 20\}$ meters.

All metrics are reported in meters. The primary results (Section 6) are computed on the held-out **test set**, with the validation set reported alongside for comparison.

4 Data Pipeline

4.1 AIS Protocol Background

The Automatic Identification System (AIS) is a maritime transponder standard mandated by the International Maritime Organization (IMO) under the SOLAS convention for all vessels above 300 gross tonnes on international voyages and all passenger ships regardless of size. Transponders broadcast vessel information over two dedicated VHF channels (161.975 MHz and 162.025 MHz) according to ITU-R Recommendation M.1371, enabling nearby vessels and shore-based receiving stations to build a real-time traffic picture without satellite relay.

Message content. Each AIS broadcast carries a combination of *static*, *voyage*, and *dynamic* data fields. The fields most relevant to trajectory prediction are summarised below:

- **MMSI** (Maritime Mobile Service Identity): a 9-digit globally unique vessel identifier, analogous to a telephone number, assigned by national maritime authorities. MMSI is the primary key used to associate observations across time.
- **Vessel name, call sign, IMO number**: static descriptors that do not change during a voyage.
- **Vessel type**: an integer code indicating the vessel category (e.g. cargo, tanker,

container, passenger, fishing, tug).

- **Position**: latitude and longitude in WGS-84 decimal degrees.
- **Speed Over Ground (SOG)**: vessel speed relative to the sea floor, in knots, with 0.1-knot resolution.
- **Course Over Ground (COG)**: direction of travel relative to true north, in degrees (0.1° resolution).
- **True Heading (HDG)**: the compass direction the bow is pointing, independent of current and wind drift.
- **Navigation Status**: a 4-bit integer encoding 16 operational states, including *under way using engine*, *at anchor*, *moored*, *engaged in fishing*, *not under command*, and *constrained by draught*. This field is self-reported by the crew and drives the transmission frequency described below.

Transmission frequency and its effect on data quality. The AIS standard defines **two transponder classes** with fundamentally different reporting rates. *Class A* transponders (mandatory on large commercial vessels) adapt their transmission interval to vessel dynamics:

- **Under way, 0–14 kn**: every 10 s; changing course: every 3.33 s.
- **Under way, 14–23 kn**: every 6 s; changing course: every 2 s.
- **Under way, >23 kn**: every 2 s.
- **At anchor or moored** ($\leq$3 kn): every 3 minutes.

Class B transponders (smaller or recreational vessels) transmit every 30 s when moving and every 3 minutes when stationary. The practical consequence is that the inter-message gap for a single vessel can range from **2 seconds** (fast-moving Class A under manoeuvre) to **3 minutes or more** (anchored or Class B), and neighbouring vessels in the same scene may each transmit at very different rates.

This variable-rate reporting is the root cause of the irregular sampling in the raw Orca AI data (typical target-vessel gaps of 20–380 s observed in practice) and motivates the interpolation and deduplication steps described in Section 4.

4.2 Raw Data Description

The dataset originates from Orca AI's proprietary AIS archive — a unique real-world collection gathered via onboard bridge transceivers installed on commercial vessels since 2019. Raw AIS streams are stored in an InfluxDB time-series database and periodically

exported as **4-hour CSV windows**, one file per ego vessel per time window. The corpus processed for this work is a subset of that archive, comprising 4-hour windows recorded across a fleet of commercial vessels.

Each CSV file encodes the ego vessel's Maritime Mobile Service Identity (MMSI), geographic position (latitude, longitude), Speed Over Ground (SOG in knots), Course Over Ground (COG in degrees), and a host identifier, alongside the states of all target vessels observed within 12 nautical miles during the same 4-hour window. The file is structured in a **multi-agent format**: each row encodes the ego vessel's state together with the state of *one* nearby target vessel at a shared timestamp. Consequently, a scene with N target vessels at a given timestamp is represented as N rows — one per target — all sharing the same ego observation.

The raw feed presents three structural challenges that the preprocessing pipeline (Section 4) addresses: **(1)** duplicate records arising from the multi-row-per-timestamp encoding; **(2)** irregular sampling intervals (AIS beacons transmit every 2–30 s depending on vessel speed and transponder class); and **(3)** geodetic coordinates that must be projected into an ego-relative Cartesian frame suitable for neural network training.

4.3 Preprocessing Steps

Step 1 — Deduplication. Two classes of duplicates are removed. First, true duplicate rows — defined as rows sharing identical values for the tuple (*timestamp, own_latitude, own_longitude, target_id*) — are dropped, retaining only the first occurrence. Second, because the multi-row-per-timestamp format causes the ego vessel's observation to appear once per target vessel, ego records are further deduplicated on (*timestamp, own_latitude, own_longitude*) before processing, ensuring each ego timestep appears exactly once.

Step 2 — Timestamp Normalization. Wall-clock timestamps are converted from ISO datetime format to floating-point seconds elapsed since the first observation in the file:

$$t_k = (T_k - T_0) \quad \text{[seconds]}.$$

This produces a scene-relative time axis starting at zero, making temporal windows invariant to absolute recording date.

Step 3 — Trajectory Interpolation. Ego vessels operate at approximately 1 Hz and require no interpolation. Target vessels exhibit irregular sampling gaps (typically 20–380 seconds). To produce a consistent 1 Hz feature stream for all agents, target trajectories are interpolated under the following policy:

- Gaps ≤ 1.5 s: no interpolation (already consistent).

- Gaps in $(1.5, 400]$ s: linear interpolation of **position only** (latitude and longitude).
- Gaps > 400 s: no interpolation; treated as a trajectory discontinuity.

SOG and COG are *not* interpolated directly, as linear interpolation of angular and speed quantities during manoeuvres produces physically unrealistic values. Instead, they are derived from the interpolated positions using finite differences:

$$\Delta x = \Delta\lambda \cdot 111320 \cdot \cos(\bar{\phi}), \quad \Delta y = \Delta\phi \cdot 110540 \tag{1}$$

$$\text{SOG} = \frac{\sqrt{\Delta x^2 + \Delta y^2}}{\Delta t \cdot 0.5144} \quad \text{[knots]} \tag{2}$$

$$\text{COG} = \left(\arctan 2(\Delta x,\ \Delta y) \cdot \frac{180}{\pi} + 360 \right) \bmod 360 \quad \text{[degrees]} \tag{3}$$

where $\bar{\phi}$ is the mean latitude of the two endpoints and Δt is the time difference in seconds. Forward differences are used at interior interpolated points; backward differences at the terminal point.

Step 4 — Coordinate Projection. All positions are converted from geodetic coordinates to a local Cartesian frame centred on a reference point $(\phi_{\text{ref}}, \lambda_{\text{ref}})$ taken as the first valid ego position in each scene:

$$x = (\lambda - \lambda_{\text{ref}}) \cdot 111320 \cdot \cos(\phi_{\text{ref}}), \quad y = (\phi - \phi_{\text{ref}}) \cdot 110540 \quad \text{[meters]}. \tag{4}$$

This flat-Earth approximation is accurate to within 0.1% for the encounter distances present in this dataset (typically < 22 km). SOG is converted to m/s and decomposed into Cartesian velocity components:

$$v_x = \text{SOG}_{\text{m/s}} \cdot \sin(\text{COG}_{\text{rad}}), \quad v_y = \text{SOG}_{\text{m/s}} \cdot \cos(\text{COG}_{\text{rad}}). \tag{5}$$

Each agent's final trajectory is stored as a matrix of shape $[T \times 5]$ with columns $[t, x, y, v_x, v_y]$.

4.4 Scene Serialization

Each processed scene is serialized as a Python pickle file containing a dictionary with the scenario identifier, per-agent state matrices (position and velocity), and the reference geodetic position for back-projection during visualization. The original geographic coordinates are preserved to enable map-based qualitative analysis.

4.5 Dataset Split

Splitting AIS data for trajectory prediction requires careful design to avoid three forms of leakage: *temporal leakage* (correlated windows from the same voyage appearing in both train and test), *vessel identity leakage* (a model memorising the behavioral signature of a specific ship seen during training), and *geographic leakage* (all test scenes drawn from a single region unseen during training). We follow a principled three-level splitting strategy that addresses all three.

Atomic unit: the 4-hour scene. Each 4-hour CSV window is treated as an indivisible unit. All 5-minute sliding sub-windows extracted from a single 4-hour scene are assigned to the same split. Splitting within a scene would place temporally adjacent, highly correlated windows into different sets, directly leaking trajectory context from train into evaluation.

Primary criterion: split by vessel MMSI. All scenes belonging to a given vessel are assigned exclusively to one split. A vessel that appears in the training set does not appear in the validation or test set. This ensures that the model is evaluated on vessels it has never encountered, measuring generalisation to unseen ships rather than memorisation of known routes and behavioral patterns.

Split ratio and reproducibility. Vessels are assigned to splits with an **80 / 10 / 10 ratio** (train / validation / test) using a fixed random seed of 42. The **same fixed split is used for all five compared models**, ensuring that performance differences reflect architectural capability rather than data partitioning. The preprocessing pipeline, model configurations and evaluation scripts are released so that the protocol can be reproduced on another corpus; the underlying AIS archive is proprietary to Orca AI and is not distributed with them.

Table 2: Data pipeline configuration and dataset statistics.

Parameter	**Value**
Input sampling rate	1–30 s (raw AIS)
Output sampling rate	1 Hz
Max interpolation gap	400 s
Coordinate frame	Ego-relative flat-Earth Cartesian
Position unit	meters
Velocity unit	m/s
Train / Val / Test split	80% / 10% / 10%



Figure 1: Validation-split overview (250 scenes, 13,820 vessel instances). From left to right: agent count per scene, speed distribution, origin-centred trajectory samples coloured by average speed, turn-rate distribution, geographic coverage, and nearest-vessel proximity (dashed lines mark 0.1/0.5/1/3 nm COLREG thresholds).

5 Methods

This section describes the five approaches compared in this work. The **Baseline Linear** model is a parameter-free least-squares extrapolator that provides a deterministic lower bound on achievable accuracy. **TrAISformer** [Nguyen and Fablet, 2021] is a GPT-style autoregressive transformer from the maritime literature that predicts trajectories as sequences of discrete tokens. **AIS-ACNet** [Shin et al., 2024] is a dilated causal CNN from the maritime literature that processes the past trajectory with exponentially-dilated convolutions. **Wayformer-AIS** is our adaptation of the Wayformer Perceiver-based architecture [Nayakanti et al., 2022], originally developed for road vehicles, which encodes multi-agent social context via cross-attention. **GAT-LSTM** [Zhao et al., 2023] is a hybrid spatial-temporal architecture combining a Graph Attention Network (GAT) for spatial feature extraction with an LSTM for temporal modelling. All models are trained and evaluated on the same data split with a fixed random seed of 42 (Section 5.6).

5.1 Baseline Linear (Least-Squares Extrapolation)

The Baseline Linear model serves as the simplest valid lower bound: it requires no training and makes no assumptions about inter-vessel interactions. Given the full sequence of past

positions $\{(x_t, y_t)\}_{t=1}^{T_{\text{past}}}$ for a vessel, the model fits a linear trend by ordinary least squares (OLS):

$$\hat{x}(t) = a_x \cdot t + b_x, \quad \hat{y}(t) = a_y \cdot t + b_y, \tag{6}$$

where the slope and intercept are estimated over all valid (unmasked) past timesteps using mean-centred regression:

$$a = \frac{\sum_{t \in \mathcal{V}} (t - \bar{t})(p_t - \bar{p})}{\sum_{t \in \mathcal{V}} (t - \bar{t})^2}, \quad b = \bar{p} - a\,\bar{t}, \tag{7}$$

where $\mathcal{V}$ is the set of valid timestep indices, $\bar{t}$ is their mean, and $\bar{p}$ is the mean position. The extrapolation to future timestep $t' \in \{T_{\text{past}} + 1, \ldots, T_{\text{past}} + T_{\text{future}}\}$ follows directly from Equation (6).

The model is implemented as a PyTorch Lightning module with a single dummy parameter for framework compatibility. It produces a single deterministic trajectory ($K = 1$); for metric consistency with $K = 6$ models, all six "modes" are set to this trajectory and $\pi_1 = 1, \pi_{2:6} = 0$. No gradient is propagated through the prediction.

The Baseline Linear is particularly informative for short horizons, where vessel motion is approximately linear. Significant degradation of its performance in dense or manoeuvring scenarios motivates the need for interaction-aware learning.

5.2 TrAISformer (Autoregressive Transformer)

TrAISformer [Nguyen and Fablet, 2021] is a GPT-style causal transformer that treats vessel trajectory prediction as a discrete sequence-to-sequence problem. Rather than regressing continuous coordinates, it quantises each timestep into a *fourhot* token: four independent categorical distributions over Δx, Δy, SOG, and COG, each discretised into a fixed vocabulary. The concatenated logit vector is fed as input to a stack of causally masked multi-head self-attention blocks, and the model is trained to predict the next token autoregressively.

Fourhot quantisation. We work in the UniTraj ego-relative coordinate frame (positions normalised by $s_p = 100\,\text{m}$, velocities by $s_v = 20\,\text{m/s}$). Empirical p99.9 ranges over the training corpus yield $x, y \in [-31, 31]\,\text{hm}$, so we allocate 256 uniform bins per spatial axis ($\approx 24\,\text{m}$ resolution), 30 SOG bins spanning 0–1 normalised speed ($\approx 0.67\,\text{m/s}$ per bin), and 72 COG bins spanning $[-\pi, \pi]$ (5°/bin, matching the original paper). The per-timestep vocabulary is $256 + 256 + 30 + 72 = 614$ logits.

Architecture. The adapted model uses $L = 8$ transformer layers, $H = 8$ attention heads, and embedding dimension $d = 256$, with dropout $p = 0.1$. A causal mask re-

stricts each position to attend only to previous tokens, enforcing a temporally consistent autoregressive factorisation. The context window spans $T_{\text{past}} + T_{\text{future}} = 600$ tokens.

Multi-modal inference. To produce $K = 6$ modes for comparison with the other models on $\text{minADE}_6/\text{minFDE}_6$, we perform K independent autoregressive rollouts with temperature sampling ($\tau = 1.0$). Modes are treated as equally probable ($\pi_k = 1/K$) since TrAISformer does not output explicit mode weights.

Training objective. The loss is the sum of four per-channel cross-entropy terms, masked to contribute only from future timesteps:

$$\mathcal{L}_{\text{TR}} = \text{CE}(\hat{q}_x, q_x) + \text{CE}(\hat{q}_y, q_y) + \text{CE}(\hat{q}_{\text{sog}}, q_{\text{sog}}) + \text{CE}(\hat{q}_{\text{cog}}, q_{\text{cog}}), \tag{8}$$

where $q_\bullet$ denotes the target bin index and $\hat{q}_\bullet$ the predicted logits.

Adaptation to the 5-minute regime. The original model was designed for 10-minute sampling intervals and 3–15 hour prediction horizons [Nguyen and Fablet, 2021]. Our regime (1 Hz, 5-min past and future) produces 600-token sequences, increasing self-attention cost approximately 31× per layer. The model operates as a single-agent (marginal) predictor: only the ego vessel's past trajectory is provided as context; surrounding vessels are ignored.

Table 3: TrAISformer hyperparameters.

Hyperparameter	**Value**
Transformer layers L	8
Attention heads H	8
Embedding dimension d	256
Dropout	0.1
Spatial bins (x, y)	256 each
SOG bins	30
COG bins	72
Modes K (inference)	6
Sample temperature τ	1.0
Optimizer	AdamW
Learning rate	6×10^{-4}
LR schedule	Cosine annealing
Gradient clip norm	1
Max epochs	100
Train batch size	8

5.3 Wayformer-AIS

Wayformer-AIS is our primary model, adapting the Wayformer architecture [Nayakanti et al., 2022] — originally developed for road-vehicle trajectory prediction on the Waymo Open Motion Dataset — to the maritime domain. The key architectural insight of Wayformer is that a Perceiver-based bottleneck encoder-decoder, using factorized attention over agents and time, achieves state-of-the-art performance with a simple, scalable design that requires no hand-crafted scene representations.

5.3.1 Input Representation and Normalization

Each agent i is represented as a sequence of 39-dimensional feature vectors over $T_{\text{past}} = 300$ timesteps. The feature vector at each timestep encodes position, velocity, heading, speed magnitude, and a binary validity flag. To bring all features into a comparable numerical range, positions are normalized by a scale of $s_p = 100\,\text{m}$ and velocities by $s_v = 20\,\text{m/s}$:

$$\tilde{x} = x/s_p, \quad \tilde{y} = y/s_p, \quad \tilde{v}_x = v_x/s_v, \quad \tilde{v}_y = v_y/s_v. \tag{9}$$

Up to $M = 10$ vessels are included per scene. Unlike road-domain applications of Wayformer, no map or lane information is used; the map input branch receives a dummy zero tensor and contributes no meaningful signal to the encoder.

5.3.2 Encoder

Agent features are first linearly projected into a $d_k = 256$ dimensional embedding space:

$$\mathbf{e}_i^t = \text{SELU}\big(\mathbf{W}_a \mathbf{x}_i^t + \mathbf{b}_a\big) + \boldsymbol{\psi}_i^{\text{agent}} + \boldsymbol{\psi}_t^{\text{time}}, \tag{10}$$

where $\boldsymbol{\psi}_i^{\text{agent}} \in \mathbb{R}^{d_k}$ and $\boldsymbol{\psi}_t^{\text{time}} \in \mathbb{R}^{d_k}$ are learnable positional embeddings for the agent index and time index respectively. The resulting agent embeddings are flattened across agents and timesteps to form a sequence $\mathbf{Z} \in \mathbb{R}^{(M \cdot T_{\text{past}}) \times d_k}$.

This sequence is passed to a **Perceiver Encoder** [Nayakanti et al., 2022] with $N_{\text{enc}} = 192$ trainable latent queries and $L_{\text{enc}} = 2$ cross-attention layers. Each cross-attention layer has $H = 4$ heads with a feedforward hidden size of 1024, followed by layer normalisation and dropout ($p = 0.1$). The cross-attention compresses the full $(M \cdot T_{\text{past}})$-length token sequence into a fixed-size context matrix $\mathbf{C} \in \mathbb{R}^{N_{\text{enc}} \times d_k}$, regardless of the number of agents or timesteps present. This bottleneck design provides robustness to variable scene sizes and missing observations.

5.3.3 Decoder

The decoder is a **Perceiver Decoder** with $N_{\text{dec}} = 128$ trainable output queries and $L_{\text{dec}} = 8$ cross-attention layers. Each output query attends to the context matrix $\mathbf{C}$ through cross-attention, followed by self-attention among the output queries themselves, producing an output sequence $\mathbf{Q} \in \mathbb{R}^{N_{\text{dec}} \times d_k}$.

The first $c = 6$ rows of $\mathbf{Q}$ correspond to the $K = 6$ prediction modes. Two lightweight heads are applied to each mode query:

$$\boldsymbol{\pi} = \text{softmax}(\mathbf{W}_p \mathbf{Q}_{1:c}) \in \mathbb{R}^c, \tag{11}$$

$$\hat{\mathbf{Y}} = \mathbf{W}_o \mathbf{Q}_{1:c} \in \mathbb{R}^{c \times T_{\text{future}} \times 5}, \tag{12}$$

where $\boldsymbol{\pi}$ are mode probabilities and $\hat{\mathbf{Y}}$ contains, for each mode k and future timestep t, the parameters of a bivariate Gaussian: $(\mu_x,\ \mu_y,\ \log \sigma_x,\ \log \sigma_y,\ \rho)$.

5.3.4 Loss Function

Training minimizes a combined regression and classification loss following the GMM formulation of Nayakanti et al. [2022]. For each sample in a batch, the mode k^* geometrically closest to the ground-truth future trajectory is selected:

$$k^* = \arg\min_k \sum_{t=1}^{T_{\text{future}}} m_t \cdot \left\| \hat{\boldsymbol{\mu}}_t^{(k)} - \mathbf{p}_t \right\|_2, \tag{13}$$

where m_t is the validity mask. The **regression loss** is the negative log-likelihood of the ground truth under the bivariate Gaussian of mode k^*:

$$\mathcal{L}_{\text{reg}} = \sum_{t=1}^{T_{\text{future}}} m_t \left[\log \sigma_x + \log \sigma_y + \tfrac{1}{2} \log(1-\rho^2) + \frac{1}{2(1-\rho^2)} \left(\frac{d_x^2}{\sigma_x^2} + \frac{d_y^2}{\sigma_y^2} - \frac{2\rho\, d_x d_y}{\sigma_x \sigma_y} \right) \right], \tag{14}$$

where $d_x = \mu_x^{(k^*)} - p_x$ and $d_y = \mu_y^{(k^*)} - p_y$ are the residuals, and $\log \sigma$ values are clipped to $[-1.609,\ 5.0]$ for numerical stability. The **classification loss** trains mode probabilities with cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\log \pi_{k^*}. \tag{15}$$

The total loss is:

$$\mathcal{L} = \frac{1}{B} \sum_{i=1}^{B} \left(\mathcal{L}_{\text{reg}}^{(i)} + \mathcal{L}_{\text{cls}}^{(i)} \right). \tag{16}$$

5.3.5 Maritime Adaptations

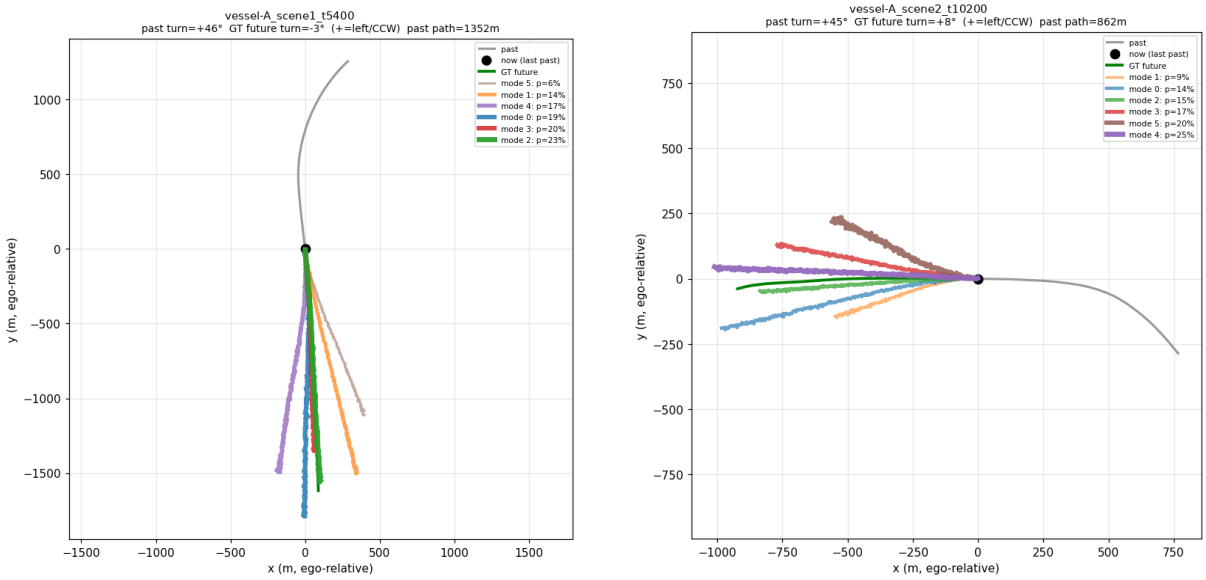
Table 4 summarises the modifications applied to the original Wayformer to accommodate the maritime domain.

Table 4: Wayformer adaptations for the maritime domain.

Aspect	**Road (original)**	**AIS (this work)**
Map input	Road network / lane polylines	None (dummy zero tensor)
Coordinate frame	World absolute	Ego-relative (meters)
Max agents	32	10
Position scale	Scene-relative	100 m
Velocity scale	km/h	m/s ($\div 20$)
Agent feature dim	39	39 (compatible)

Figures 2 and 3 present two qualitatively distinct validation scenarios that illustrate the multi-modal output of Wayformer-AIS and its ability to capture different maritime motion regimes.

Scenario 1 — Active manoeuvre (Figure 2). Vessel A is observed completing a $+45°$ port turn over an 862 m past path. The model produces $K = 6$ candidate future trajectories, each with a distinct softmax probability π_k. The highest-weighted mode (mode 4, $p = 25\%$, purple) predicts a near-straight continuation consistent with the vessel easing out of the turn, while lower-probability modes span alternative headings — port deviations, starboard corrections, and wider arcs. The ground-truth future (green, $+8°$ port) falls squarely within the predicted fan, confirming that the model correctly frames an ongoing manoeuvre as a *distribution* over plausible future headings. Two consecutive timestamps of the same vessel are shown, both exhibiting consistent multi-modal spread across the turn. This probabilistic coverage is the key structural advantage of Wayformer-AIS over the single-mode models (AIS-ACNet, GAT-LSTM) and directly enables the Brier-FDE calibration metric.



(a) Offset $t = 5400\,\mathrm{s}$ (b) Offset $t = 10200\,\mathrm{s}$

Figure 2: Wayformer-AIS predictions for Vessel A (2024-01-01) at two timestamps during an active port-turn manoeuvre (past path = 862 m, past turn = +45° CCW, GT future turn = +8°). **Gray**: past trajectory. **Black dot**: current position (ego-relative origin). **Green**: ground-truth future. **Coloured**: six predicted modes ($p = 9\%$–25%). The fan of modes across port and starboard headings reflects genuine uncertainty in the vessel's intended manoeuvre.

Scenario 2 — Deceleration and anchoring (Figure 3). Vessel B approaches its anchor point after a −156° heading change over 332 m of past travel. The critical signal is in the spacing of position fixes in the final seconds of the past window: the AIS points crowd tightly together near the current position, a direct kinematic signature of rapid deceleration at 1 Hz resolution. Wayformer-AIS recognises this pattern and responds with a striking prediction: *all six modes cluster within approximately 20 m of the current position*, and the dominant mode (mode 3, $p = 35\%$) predicts essentially zero future displacement. The model has inferred, purely from velocity and position features in the AIS stream, that the vessel is coming to rest. The ground-truth future (green) confirms this — the vessel anchors and remains nearly stationary for the entire 300 s forecast horizon. Note that the axis range in Figure 3 is only ±50 m, an order of magnitude smaller than the manoeuvring scenario above, which visually underscores how tightly the model constrains the forecast when it detects a stopping vessel. This example demonstrates that Wayformer-AIS captures not only directional uncertainty but also speed dynamics, distinguishing cruising, manoeuvring, and anchoring regimes without any explicit navigation-status input.

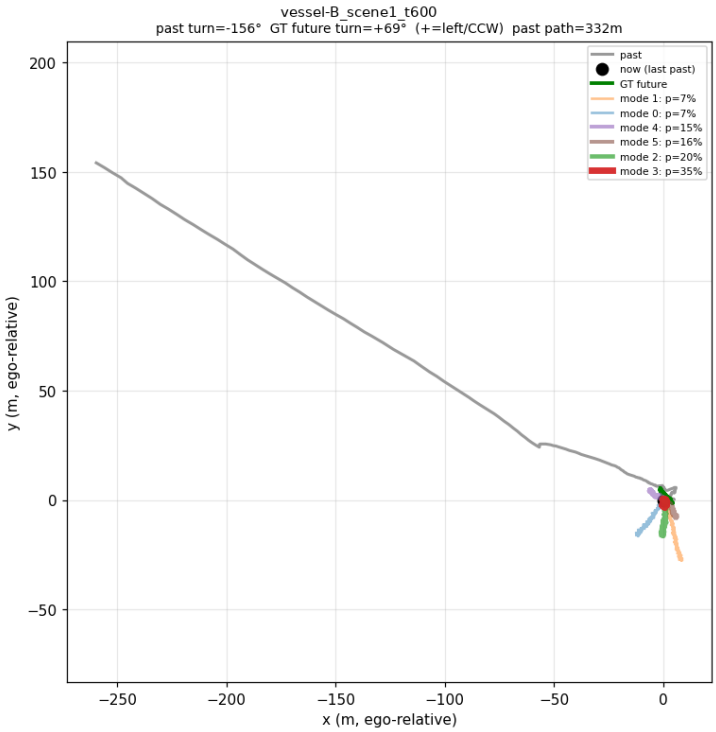


Figure 3: Wayformer-AIS prediction for Vessel B (2024-01-04) approaching anchor (past path = 332 m, past turn = −156°). **Gray**: past trajectory — note the tightly-spaced position fixes near the current position, signalling rapid deceleration. **Green**: ground-truth future, confirming near-zero displacement over the 300 s horizon. **Coloured**: all six predicted modes ($p = 7\%$–35%) cluster within ≈ 20 m of the origin. The axis range (± 50 m) is an order of magnitude smaller than Figure 2, highlighting the model's ability to predict an imminent anchoring event from kinematic features alone.

5.4 AIS-ACNet (Dilated Causal CNN)

AIS-ACNet [Shin et al., 2024] predicts vessel trajectories using a *dual-encoder dilated causal convolutional network* (GWNet backbone). The architecture processes the past trajectory sequence with two parallel 1-D dilated causal CNN stacks — one for position (x, y) and one for kinematics (SOG, COG) — whose intermediate activations are fused via gated residual connections at every layer.

Exponential dilations. The network stacks $N = 10$ dilated causal convolution layers with exponential dilation factors $[1, 2, 4, 8, 16, 32, 64, 128, 256, 512]$ and kernel size $k = 2$. The theoretical receptive field is $1 + (k-1)\sum_{i=1}^{N} d_i = 1 + 1023 = 1024$ timesteps, fully covering the 300-step past horizon with margin. The original paper used a smaller dilation schedule designed for its 15-step window; our exponential schedule is the key adaptation for the 5-minute / 1 Hz regime.

Architecture. Each layer has residual channels $C_r = 32$ and dilation channels $C_d = 32$; skip connections from all layers are summed into a skip aggregation with $C_s = 256$

channels, followed by a two-layer output head with $C_e = 512$ intermediate channels. Total trainable parameters: approximately 1.2 M. The model is single-mode (deterministic): it outputs one future trajectory ($K = 1$), so all six "modes" are identical when computing minADE_6 and minFDE_6.

Training objective. Following the original paper, the loss combines MSE on future positions with two auxiliary SOG and COG regression terms:

$$\mathcal{L}_{\text{ACN}} = \mathcal{L}_{\text{pos}} + \alpha\,\mathcal{L}_{\text{SOG}} + \beta\,\mathcal{L}_{\text{COG}}, \quad \alpha = \beta = 0.2. \tag{17}$$

Table 5: AIS-ACNet hyperparameters.

Hyperparameter	**Value**
Dilation factors	$[1, 2, 4, 8, 16, 32, 64, 128, 256, 512]$
Residual / dilation channels C_r	32
Skip channels C_s	256
Output head channels C_e	512
Kernel size	2
Dropout	0.3
Modes K	1 (deterministic)
Auxiliary loss weights α, β	0.2
Optimizer	Adam
Learning rate	10^{-3}
LR decay epochs	80, 90
Weight decay	10^{-4}
Gradient clip norm	5
Max epochs	100
Train batch size	32

5.5 GAT-LSTM (Graph Attention Network + LSTM)

GAT-LSTM [Zhao et al., 2023] combines a Graph Attention Network (GAT) for spatial feature extraction with a Long Short-Term Memory (LSTM) network for temporal modelling. The method was proposed by Zhao et al. [2023] for ship trajectory prediction on Chinese coastal AIS data; we adapt it here to the 5-minute horizon used throughout this benchmark.

Graph construction. The ego vessel's past trajectory is partitioned into contiguous sub-segments, each forming a node in a directed graph $G = (P, E)$. Two nodes i and j

are adjacent when their sub-segments are temporally consecutive, yielding an adjacency matrix $A \in \{0,1\}^{P\times P}$ with a self-loop added to every node: $\tilde{A} = A + I$.

Spatial encoder (GAT). At each timestep the GAT layer computes attention coefficients between node i and its neighbours $\mathcal{N}(i)$:

$$\alpha_{ij} = \frac{\exp\big(\mathrm{LeakyReLU}(e_{ij})\big)}{\sum_{k\in\mathcal{N}(i)} \exp\big(\mathrm{LeakyReLU}(e_{ik})\big)}, \quad e_{ij} = a\big(Wh_i^l,\ Wh_j^l\big)\,, \tag{18}$$

where $h_i^l \in \mathbb{R}^F$ is the node-i feature at GAT layer l, $W \in \mathbb{R}^{F'\times F}$ is a learnable weight matrix, and $a(\cdot)$ is a shared attention function. Updated node features are:

$$h_i^{l+1} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}\, W h_j^l\right). \tag{19}$$

Multi-head attention with $K_{\text{heads}} = 4$ independent mechanisms is applied; their outputs are concatenated before being passed to the LSTM.

Temporal decoder (LSTM). The GAT-encoded feature at each past timestep is fed sequentially into a 2-layer LSTM with hidden size 128. The final hidden state $h_{T_{\text{past}}}$ is projected by a linear layer to a flat vector, then reshaped to $T_{\text{future}} \times 2$ predicted (x, y) positions. The model is deterministic ($K = 1$); for metric consistency, all six modes are set to this single trajectory with $\pi_1 = 1$.

Adaptation to the 5-minute regime. The original architecture was designed for short prediction horizons (10 s – 1 min) with next-step autoregressive decoding. Our adaptation replaces the autoregressive decoder with a direct multi-step linear projection from the LSTM final hidden state, predicting all $T_{\text{future}} = 300$ future positions at once.

Table 6: GAT-LSTM hyperparameters.

Hyperparameter	**Value**
GAT hidden size	32
GAT layers	2
Attention heads K_{heads}	4
LSTM hidden size	128
LSTM layers	2
Input features	4 (x, y, v_x, v_y)
Dropout	0.3
Modes K	1 (deterministic)
Optimizer	Adam
Learning rate	10^{-3}
LR decay epochs	80, 90
Weight decay	10^{-4}
Gradient clip norm	5
Max epochs	100
Train batch size	32

5.6 Shared Training Protocol

All learned models are trained under the following shared conditions to ensure that performance differences reflect architectural choices rather than experimental setup.

Data. All models are trained and evaluated on the same fixed data split, constructed according to the procedure described in Section 4: each 4-hour scene is assigned whole to a single split, so that no observation window is shared between partitions. A fixed random seed of 42 is used throughout, and the preprocessing pipeline and normalisation constants ($s_p = 100\,\text{m}$, $s_v = 20\,\text{m/s}$) are identical across models.

Uniform evaluation protocol. The results in Section 6 are produced by re-evaluating every model under a single common protocol, rather than by collating the metrics each model happened to log during its own training run. All five models are scored on the complete split with no per-run cap on the number of scenes loaded, using the same observation and prediction horizons ($T_{\text{past}} = T_{\text{future}} = 300\,\text{s}$ at $1\,\text{Hz}$), the same $300\,\text{s}$ sliding-window stride, the same number of output modes ($K = 6$), and the same metric implementations. This yields 11,891 prediction windows on the test split and 10,336 on the validation split, and within a split every model is scored on exactly the same windows.

Enforcing this uniformity is essential to the comparison. Metrics recorded during separate

training runs are not directly comparable, because those runs may differ in how many scenes were loaded and in the stride used to extract windows; a difference between two such figures reflects the evaluation sample as much as the architecture. Model selection and reporting are also kept apart: each checkpoint is chosen on the validation split (see below), while the primary results are reported on the held-out test split.

Checkpoint selection. The best model checkpoint is selected by minimising the validation Brier-FDE over all training epochs, where Brier-FDE [Feng et al., 2024] is defined as:

$$\text{Brier-FDE} = \text{minFDE}_K + (1 - \pi_{k^*})^2, \tag{20}$$

and π_{k^*} is the predicted probability of the best mode. This metric jointly rewards spatial accuracy and calibrated confidence.

Hardware. All learned models are trained on an AWS EC2 `g5.8xlarge` instance equipped with a single NVIDIA A10G GPU (24 GB VRAM), an AMD EPYC 7R32 CPU (32 vCPUs), and 128 GB of system RAM. The Baseline Linear requires no GPU and is evaluated on CPU.

Experiment tracking. All training runs and evaluation results across every architecture are logged to a shared project on the Weights & Biases (W&B) platform [Biewald, 2020]. Each run records per-epoch training and validation losses, the four evaluation metrics, gradient norms, and hardware utilisation, providing a single point of visibility for cross-model comparison. This centralised tracking was instrumental in identifying the best checkpoint for each model and in reproducing any reported result from its exact hyperparameter configuration.

Per-model training details. Each model uses its own optimizer and schedule, reflecting best practices from its source paper. Table 7 summarises the key differences.

Table 7: Per-model training configuration. All models use the same data split, seed (42), and checkpoint selection criterion (val/brier_fde).

Model	Optimizer	LR / Schedule	Epochs	Best ckpt epoch
Baseline Linear	—	—	—	—
TrAISformer	AdamW	6×10^{-4} / cosine	100	32
AIS-ACNet	Adam	10^{-3} / step at [80,90]	100	99
Wayformer-AIS	AdamW	10^{-3} / OneCycleLR	100	96
GAT-LSTM	Adam	10^{-3} / step at [80,90]	100	20

6 Experiments

6.1 Quantitative Comparison

Table 8: Main results under an identical evaluation protocol for every model ($T_{\text{past}} = T_{\text{future}} = 300\,\text{s}$, $K = 6$, window stride 300 s, all scenes in the split, seed 42). Upper panel: held-out test set. Lower panel: validation set. ↓ = lower is better; best learned model in **bold**. The parameter-free OLS baseline is the reference line — learned models that fall below it do not beat constant-velocity extrapolation.

Model	**minADE$_6$** ↓ (m)	**minFDE$_6$** ↓ (m)	**Brier-FDE** ↓	**Miss@2 m** ↓ (%)	**Miss@10 m** ↓ (%)	**Miss@20 m** ↓ (%)
Test set — 11,891 prediction windows						
Baseline Linear (OLS)	26.70	51.43	51.43	95.3	68.1	45.8
Wayformer-AIS	**15.57**	**31.71**	**32.31**	**91.6**	**67.3**	**41.6**
AIS-ACNet†	27.03	56.76	56.76	96.4	78.8	65.7
TrAISformer	100.43	186.49	187.19	99.8	81.9	73.5
GAT-LSTM†	409.42	819.12	819.12	100.0	100.0	99.7
Validation set — 10,336 prediction windows						
Baseline Linear (OLS)	34.19	65.85	65.85	95.7	71.3	49.9
Wayformer-AIS	**18.07**	**36.94**	**37.56**	**92.7**	**71.3**	**44.2**
AIS-ACNet†	33.13	69.90	69.90	96.2	81.1	67.6
TrAISformer	101.19	188.97	189.66	99.6	85.0	77.6
GAT-LSTM†	379.23	756.05	756.05	100.0	100.0	99.5

† Single-mode by architecture: minADE$_6$ is effectively minADE$_1$ and Brier-FDE collapses onto minFDE$_6$.

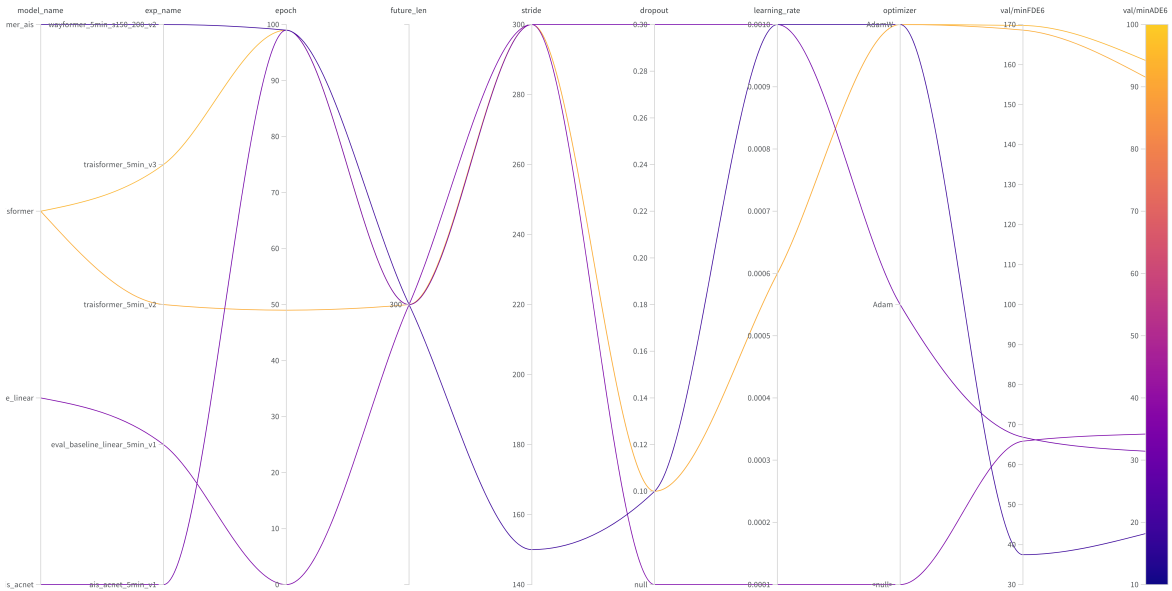


Figure 4: Parallel coordinates view of all model runs on the W&B experiment tracking platform [Biewald, 2020]. Each line represents one run; axes show key hyperparameters (epoch, future horizon, stride, dropout, learning rate, optimiser) and final validation metrics ($val/minFDE_6$, $val/minADE_6$). Lines are colour-coded by $val/minADE_6$ (dark purple → low/good, orange → high/poor). Wayformer-AIS (bottom-right cluster, dark purple) achieves the lowest minADE_6 across all runs; GAT-LSTM and TrAISformer runs trail significantly in both metrics.

6.2 Performance by Prediction Horizon

A single aggregate metric such as minADE_6 collapses all 300 future timesteps into one number and can obscure where in the prediction window each model breaks down. Figure 5 decomposes that number: the prediction horizon is truncated at each of the 300 timesteps in turn and minADE_6 recomputed there, for all five models on the held-out test split. Every point on every curve is therefore measured rather than extrapolated, and the value each curve takes at 300 s reproduces that model's row in Table 8. The axis stops at five minutes because the evaluation window itself is 300 s long: beyond it there is no ground truth to measure against, and every learned output head in the comparison is exactly 300 timesteps wide. A logarithmic ordinate is necessary to accommodate GAT-LSTM's error of several hundred metres alongside the other models.

Several trends emerge from the figure. First, **Wayformer-AIS** holds the lowest error at effectively every point in the window, from 3.67 m at one minute to 15.57 m at five, against 9.77 m and 26.70 m for the Baseline Linear model — a 1.71× improvement at the evaluation horizon and a wider one earlier. Second, **TrAISformer** is worse than constant-velocity extrapolation at every timestep without exception, and falls further behind as the horizon lengthens: its autoregressive generation accumulates error at each decoding

step, reaching 100.43 m at five minutes, nearly four times the Baseline. This behaviour is consistent with the known compounding-error pathology of autoregressive sequence models applied to regression tasks [Nguyen and Fablet, 2021].

Third, **AIS-ACNet** shows that the ranking at five minutes is not the ranking throughout. It is more accurate than the Baseline Linear model for the first 281 seconds of the window — 5.84 m against 9.77 m at one minute, a 40 % reduction — and surrenders that advantage only in the final 19 seconds, finishing marginally behind on minADE_6 (27.03 m against 26.70 m) and more clearly behind on minFDE_6 (56.76 m against 51.43 m), the metric that scores the endpoint alone. Its error grows faster than the kinematic reference even while it remains below it, so the dilated causal convolution buys a real but temporary advantage, one that has expired by the horizon at which the comparison is scored. This is the decomposition earning its place: an aggregate at 300 s records AIS-ACNet as a model that does not beat constant velocity, while the curve shows it beating constant velocity for 94 % of the window and losing on the approach to the endpoint. Against that, Wayformer-AIS is the only model whose margin over the baseline is sustained from the first timestep to the last.

Finally, **GAT-LSTM** diverges from the outset, reaching 409.42 m minADE_6 by five minutes — a Miss Rate of 100 % at the 2 m and 10 m thresholds confirms the model fails to produce any usable prediction on this dataset, likely because its single-step ADE training objective does not generalise to 300-step rollouts at 1 Hz maritime resolution.

Read against the OLS reference line, these results carry a sharper message than a simple ranking. Of the four learned architectures, *only* Wayformer-AIS beats constant-velocity extrapolation at the five-minute evaluation horizon; AIS-ACNet, TrAISformer and GAT-LSTM all fall short of it there, on both splits. That a dilated convolution, an autoregressive transformer and a graph-recurrent hybrid each lose to a parameter-free linear fit is the most consequential outcome of this comparison: at a five-minute maritime horizon, learned sequence modelling does not by itself improve on constant-velocity extrapolation, and a study reporting only learned models against one another would have concealed that.

The architectures also differ in what they are permitted to observe, and on this axis the field divides cleanly. AIS-ACNet, TrAISformer and GAT-LSTM are all *single-vessel* predictors: each forecasts a ship from its own past track, and the presence, bearing and motion of every other vessel in the scene are invisible to them. Wayformer-AIS alone is given the surrounding traffic as input, attending over as many as ten vessels in the encounter and forming its prediction in the context of the scene rather than of one isolated track. It is, in other words, the only model in this comparison equipped to treat a manoeuvre as a response to anything.

It is also the only one that beats constant-velocity extrapolation at the evaluation horizon,

and the only one that does so at every point before it. The two facts sit together and are, at minimum, mutually consistent: the architecture with access to scene context is the architecture that clears the kinematic reference line, and it does so by the widest margin in exactly the regimes where a vessel's motion is most plausibly shaped by the vessels around it (Section 6.3). We state the association rather than a mechanism, since interaction-awareness and the winning architecture coincide in a single model and the margin in Table 8 reflects that architecture as a whole. What the comparison does establish is narrower but not weak: among the designs examined here, the one built to perceive its surroundings is the only one whose predictions are worth preferring to a straight line.

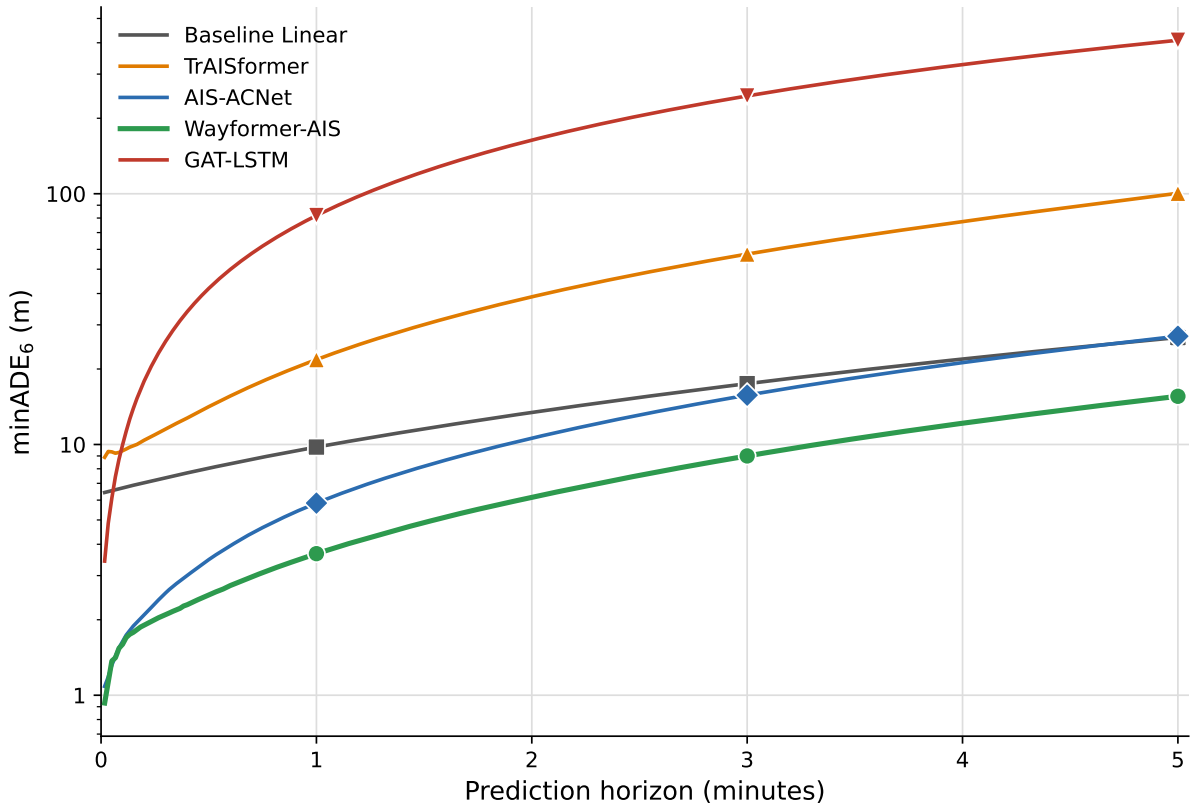


Figure 5: minADE$_6$ against prediction horizon for all five models on the held-out test split (log scale). The horizon is truncated at each of the 300 timesteps in the window and the metric recomputed there, so every point is measured rather than extrapolated; markers highlight 1, 3 and 5 minutes, and the value at 5 minutes reproduces each model's row in Table 8. Wayformer-AIS holds the lowest error across the window; AIS-ACNet runs below the Baseline Linear model for the first 281 s before crossing above it; GAT-LSTM diverges from the outset (minADE$_6$ = 409.42 m at 5 min, Miss@2 m = 100 %).

6.3 Performance by Manoeuvre Regime

Aggregate metrics average over every kind of vessel behaviour, from a ship at anchor to one executing a hard course alteration. Because these regimes differ by more than an

order of magnitude in intrinsic difficulty, an aggregate figure can conceal both where a model earns its advantage and where it fails. We therefore stratify the test set by the *ground-truth manoeuvre* performed during the prediction window, classifying each window by the total heading change of the observed future trajectory: **stationary** (displacement below the AIS noise floor), **straight** ($< 10°$), **mild turn** ($10°$–$30°$) and **sharp turn** ($\geq 30°$). Table 9 reports minFDE_6 within each regime for all five models, together with each model's margin against the OLS baseline evaluated on the identical windows.

Table 9: Test-set minFDE_6 (m) by ground-truth manoeuvre regime, with each model's margin against the OLS baseline on the same windows (Δ; negative = better than constant-velocity extrapolation). Bucket shares are of the 11,891 test windows.

		Wayformer-AIS		**AIS-ACNet**		**TrAISformer**		**GAT-LSTM**	
Regime	**OLS**	minFDE	Δ	minFDE	Δ	minFDE	Δ	minFDE	Δ
Stationary (18.5 %)	6.05	**2.84**	−3.20	6.18	+0.13	8.12	+2.08	648.57	+642.52
Straight (13.6 %)	35.26	**22.25**	−13.00	48.52	+13.26	181.13	+145.88	859.06	+823.80
Mild turn (60.9 %)	61.31	**38.44**	−22.87	70.37	+9.07	237.20	+175.89	866.01	+804.71
Sharp turn (6.9 %)	117.71	**68.25**	−49.46	88.53	−29.18	228.27	+110.56	784.59	+666.88

The advantage concentrates where extrapolation fails. Wayformer-AIS is the only model that improves on the OLS baseline in *every* regime, and the size of that improvement grows monotonically with manoeuvre difficulty: −3.20 m on stationary vessels, −13.00 m on straight passages, −22.87 m on mild turns and −49.46 m on sharp turns. The advantage is therefore not distributed evenly across the split but concentrates in the windows where constant-velocity extrapolation is weakest. A model that merely reproduced the vessel's recent velocity would track the baseline everywhere and show a flat margin; the observed gradient shows instead that Wayformer-AIS retains useful accuracy precisely where the kinematic assumption fails, which is the regime that matters operationally, since it is the manoeuvring vessel rather than the steady one that drives a collision-avoidance decision.

The competing architectures fail in diagnostically different ways. AIS-ACNet is *worse* than constant velocity on the easy regimes (+13.26 m on straight passages, +9.07 m on mild turns) while beating it on sharp turns (−29.18 m): its dilated convolutions capture some curvature but at the cost of injecting spurious motion into windows that a straight-line extrapolation would have handled correctly. TrAISformer loses to the baseline in every regime, by margins (+110 to +176 m) far exceeding the errors being measured, consistent with the compounding-error behaviour identified in Section 6.2. GAT-LSTM's stationary-regime minFDE_6 of 648.57 m is the most informative single number in the table: a model that forecasts more than half a kilometre of displacement for a vessel that does

not move has not learned the task in any usable sense, and its apparently "better" relative performance on sharp turns reflects only that the reference errors are larger there.

A caveat on the bucket boundaries. The mild-turn regime accounts for 60.9 % of test windows, which is large for a category defined by a 10° threshold. Part of this mass reflects genuine low-rate course adjustment, and part reflects heading jitter in the AIS signal at 1 Hz resolution; the stationary bucket is separated first precisely because reported heading is unreliable at very low speed. The stratification should therefore be read as a difficulty ordering rather than as a navigationally exact taxonomy.

6.4 Qualitative Analysis

We examine two representative scenarios from the validation set to illustrate the behaviour that the aggregate metrics capture only partially.

Scene 1 — Dense traffic in the Yellow Sea. Figure 6 shows a ten-vessel scene recorded off the west coast of South Korea on 23 September 2024. Vessel 0 (the prediction target) is a large cargo vessel navigating through a congested shipping lane. The model produces six hypothesis trajectories; the best mode closely follows the ground-truth course alteration, yielding a per-scene $\text{minADE}_6 = 14.57\,\text{m}$, below the dataset mean of 18.1 m. A kinematic baseline that projects the most recent heading forward would miss the turn entirely. This demonstrates that the cross-vessel attention in Wayformer-AIS extracts actionable social context: the presence and relative bearing of nine surrounding vessels provides a latent signal that a course change is imminent, consistent with COLREG Rule 16 (give-way vessel obligation).

Scene 2 — Stationary vessel during pilot transfer. Figure 7 shows a qualitatively distinct scenario from 18 November 2024. Vessel 0 is stationary — likely waiting at the pilot boarding ground outside a port while a second vessel approaches and comes alongside to transfer the harbour pilot. Despite the strong social signal from an approaching vessel that would, in most contexts, trigger an evasive prediction, Wayformer-AIS correctly predicts near-zero displacement for the target. This is notable: the model has implicitly learned from training data that a stationary vessel in close proximity to a small approaching boat does not manoeuvre — a maritime convention with no analogue in road-traffic datasets. The scenario represents an out-of-distribution event that the model handles without incident: the observed kinematics of both vessels — one stationary, one closing slowly — are consistent with a controlled approach rather than a developing encounter.

Failure modes. Both scenes above are success cases, selected to show what the model does when it works. They are not representative: the miss rate of 91.6 % at the 2 m

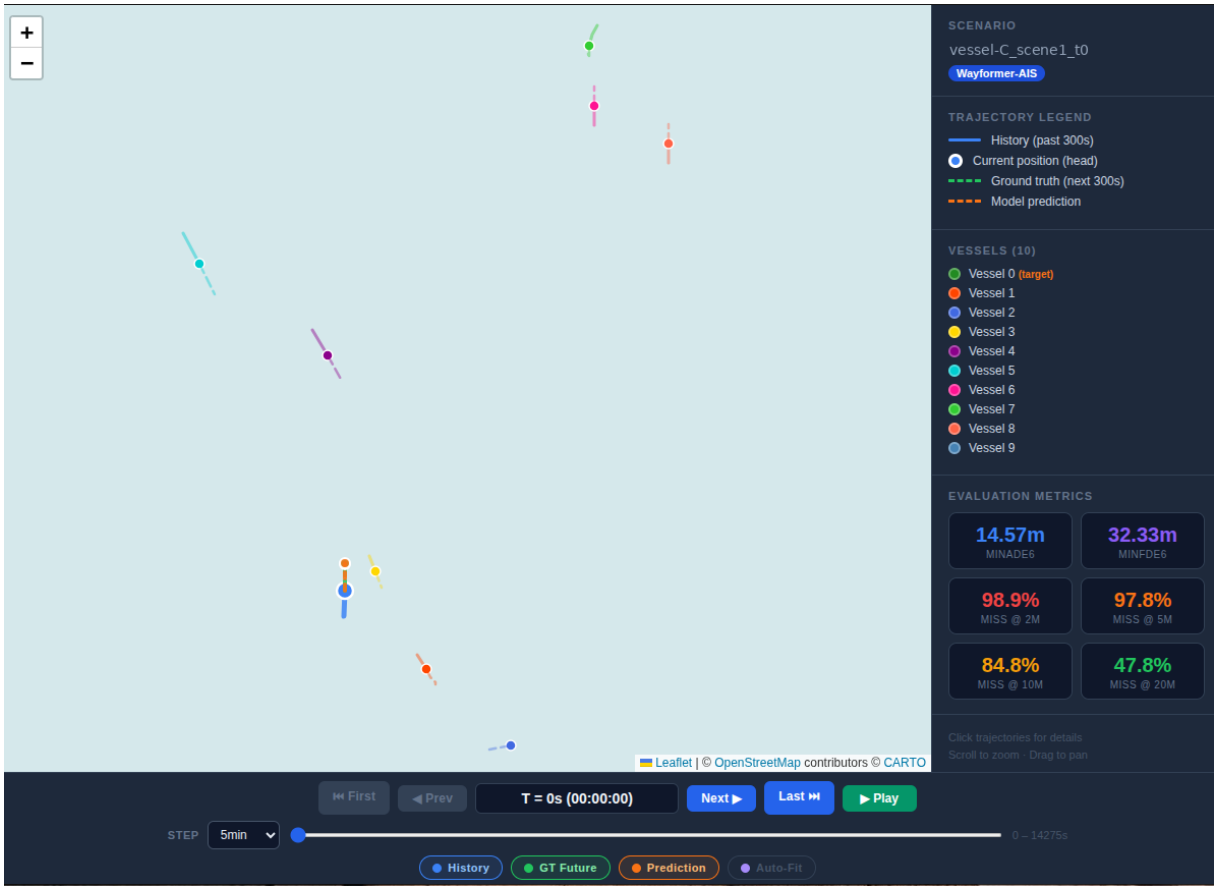


Figure 6: Dense ten-vessel scene in the Yellow Sea (23 Sep 2024, scene `vessel-C_scene1`). Solid lines show the 300 s observation history; green dashed line is the ground-truth future; orange dashed lines are the six Wayformer-AIS mode predictions. The best mode captures the vessel's course alteration in a congested COLREG environment. $\text{minADE}_6 = 14.57\,\text{m}$, $\text{minFDE}_6 = 32.33\,\text{m}$.

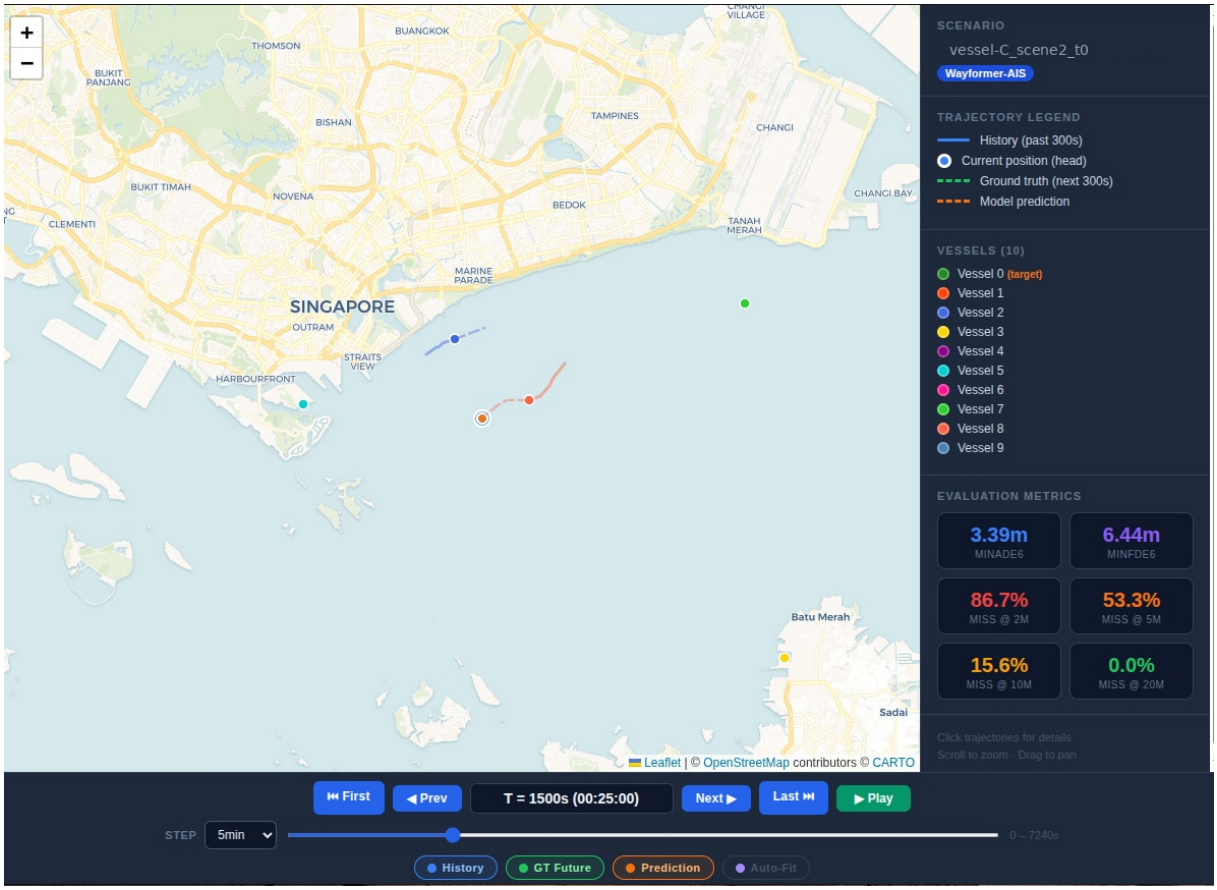


Figure 7: Singapore Strait scenario (18 Nov 2024, scene `vessel-C_scene2`). Vessel 0 (target, green) is nearly stationary in the strait south of Singapore while Vessel 1 (orange) approaches from the east. Wayformer-AIS predicts near-zero displacement, consistent with the vessel awaiting a pilot or holding position before port entry. All six prediction modes remain close to the observed position ($\text{minADE}_6 = 3.39\,\text{m}$, $\text{minFDE}_6 = 6.44\,\text{m}$).

threshold on the test set makes clear that most windows are not predicted to that tolerance. Section 6.5 characterises the failures systematically rather than anecdotally.

6.5 Failure Modes

An aggregate error figure reports how far a prediction lands from the truth, but not what kind of mistake produced that distance. This section characterises the failures of Wayformer-AIS on the held-out test split.

A turn already in progress is predicted as a straight line. Restricting attention to *moving* windows whose observed past curves by at least 15° — 156 of the 11,891 test windows, or 1.3 % — the ground truth continues the turn in 46.2 % of cases and straightens in 46.8 %, close to an even split. Wayformer-AIS does not reproduce that ambiguity: its highest-ranked mode predicts a straight continuation in 98.1 % of these windows. Narrowing to the 72 windows in which the vessel was turning *and* went on turning, the top mode straightens 71 times (Table 10). The limitation is shared rather than particular to this model — the single-vessel AIS-ACNet straightens in all 72 — and the one model that does continue turns, TrAISformer, attaches those better-oriented predictions to trajectories several hundred metres from the truth (sharp-turn minFDE_6 228.27 m against 68.25 m, Table 9).

Table 10: Behaviour when the observed past is curved and the vessel continues to turn ($n = 72$ test windows). A model reproducing the ground-truth distribution would continue the turn in roughly half of these cases.

	Continues the turn	**Straightens**
Ground truth	46.2 %	46.8 %
Wayformer-AIS (top mode)	0.6 %	**98.6 %**
AIS-ACNet	0.0 %	100.0 %
TrAISformer	16.7 %	56.9 %

The error is not a bias toward straight lines. It would be natural to read that result as the model defaulting to straight-line motion, but the aggregate statistics contradict it. The mean ground-truth heading change across the test split is 12.7°, while the mean heading change of the top-ranked mode is 33.7°, more than twice as large. Sharp turns are emitted in 10.1 % of windows against a true incidence of 6.9 %, and straight passages in 8.4 % against a true 13.6 %. The model therefore turns *more* than the fleet does in aggregate while failing to turn where a turn is already underway, which is consistent only if turning is allocated to the wrong windows — a fault in the ranking of hypotheses rather than in their generation. Section 7.1 examines that distinction directly.

Failures concentrate on fast vessels in dense traffic. Errors are smallest for vessels at anchor (minADE$_6$ 3.17 m) and largest in the 4–10 kn band (26.22 m) — the sparse middle of this fleet's bimodal speed distribution, where a vessel is neither stopped nor settled on passage but accelerating, decelerating or entering port — with vessels at high speed next at 22.00 m. Once speed is held fixed, additional traffic makes prediction harder rather than easier: for vessels at service speed minFDE$_6$ rises from 30.5 m with no neighbour in view to 68.8 m with six or more. The hardest windows are therefore underway vessels in company — precisely the situations in which a prediction would inform a collision-avoidance decision.

7 Discussion

7.1 What the reported metrics do and do not measure

The manoeuvre statistics in Section 6.5 point at the ranking of predicted hypotheses rather than their generation. That distinction can be tested directly. minADE$_6$ and minFDE$_6$ select the closest of the six predicted modes *after* comparing them with the ground truth; no deployed system can perform that selection, and must instead act on the mode the model ranks highest. Table 11 separates the two.

Table 11: Oracle versus committed prediction on the test split. *Oracle* = the best of the six predicted modes, selected after comparison with the ground truth (the standard minADE$_6$ / minFDE$_6$ metrics). *Committed* = the single highest-probability mode, i.e. the trajectory a deployed system would actually act on. The three single-mode models emit one trajectory, so the two columns coincide for them by construction.

	Oracle (best of six)		**Committed (top-ranked mode)**		
Model	ADE (m)	FDE (m)	ADE (m)	FDE (m)	Miss@10 m
Wayformer-AIS	**15.57**	**31.71**	**25.98**	54.54	77.3 %
Baseline Linear (OLS)	26.70	51.43	26.70	**51.43**	**68.1 %**
AIS-ACNet	27.03	56.76	27.03	56.76	78.8 %
TrAISformer	100.43	186.49	168.64	377.72	84.5 %
GAT-LSTM	409.42	819.12	409.42	819.12	100.0 %

For Wayformer-AIS the gap between the two settings is 10.41 m of ADE and 22.83 m of FDE, roughly 40 % of the reported figure in each case. Two conclusions follow. First, the hypothesis set is sound: a trajectory close to the truth is present among the six far more often than the committed prediction suggests, so the model is capable of representing the manoeuvre it declines to commit to. Second, the probability head orders those hypotheses poorly, and once a single trajectory must be chosen the terminal error approaches

that of constant-velocity extrapolation, although the committed ADE remains the lowest of the five models. Among the two multi-modal architectures Wayformer-AIS is the better calibrated — its FDE inflates by a factor of 1.72 without the oracle against 2.03 for TrAISformer, whose equiprobable modes make its top-ranked trajectory an arbitrary selection rather than a ranked one. The multi-modal output is therefore doing real work, but its value depends on a consumer able to weigh several hypotheses, such as a planner testing candidate futures for clearance. Even under the oracle metric a 10 m terminal error is exceeded in 67.3 % of windows, rising to 77.3 % for the committed prediction: at a five-minute horizon the task is not solved by any model examined here.

7.2 A learned model is not automatically better than a kinematic one

The most consequential outcome of this comparison is not which architecture ranked first but how many ranked below the reference line. Three of the four learned models — a dilated causal convolution, an autoregressive transformer and a graph-recurrent hybrid — produce larger terminal errors than ordinary least-squares extrapolation of the observed track (Table 8). Each was published as an advance in vessel trajectory prediction, and each was re-implemented here with the hyperparameters its authors reported. Evaluated against one another they would form a credible ranking; evaluated against a parameter-free linear fit, three of them fail. A comparison that omits the kinematic baseline — as the maritime prediction literature surveyed in Section 2 generally does — cannot distinguish a model that has learned vessel behaviour from one that has learned less than a straight line.

Part of the explanation lies in the traffic itself. The fleet in this corpus is strongly bimodal in speed (Figure 1): 23.6 % of test prediction windows are below 1 knot and 51.1 % fall between 10 and 16 knots, with very little mass in between — only 3.6 % of windows lie in the 4–10 knot band. Vessels are therefore either stationary — at anchor, alongside, or holding position — or proceeding at service speed on a settled heading. Both regimes are modelled almost exactly by constant-velocity extrapolation, which is why that baseline is far stronger here than its simplicity suggests, and why the manoeuvring minority (Section 6.3) carries nearly all of the discriminating power in the comparison. It is in that minority that Wayformer-AIS separates itself: its margin over the baseline widens monotonically from −3.20 m on stationary vessels to −49.46 m on sharp turns.

7.3 Where prediction remains hardest

The conditions under which these models struggle are not the ones an aggregate figure suggests. Vessels at anchor are predicted most accurately of all (minADE_6 3.17 m), and

error tracks what the vessel is doing rather than how close other traffic is: the hardest regime is not the fastest but the 4–10 kn band (26.22 m), where vessels are changing state rather than holding one. Once speed is held fixed, additional vessels in the scene make prediction harder: for ships at service speed, minFDE_6 rises from 30.5 m with no neighbour in view to 68.8 m with six or more (Section 6.5). The hardest windows in the dataset are therefore underway vessels in company, which is precisely the situation in which a forecast would inform a collision-avoidance decision, and precisely the situation the literature's single-vessel formulations are least equipped to represent.

The most specific failure is temporal rather than spatial. When a vessel is already mid-manoeuvre, the ground truth continues the turn about as often as it straightens, but the highest-ranked predicted mode straightens in 98.1 % of such windows. Read together with Section 7.1, this is better understood as a ranking failure than as an inability to represent the manoeuvre: the hypothesis set usually contains a plausible continuation, and the probability head does not place it first.

7.4 Limitations

Three limitations bound the conclusions that can be drawn. The data come from a single operator's fleet of commercial vessels; the traffic patterns, vessel classes and operating areas it covers are not a random sample of world shipping, and the speed distribution described above is a property of this fleet rather than of maritime traffic in general. The manoeuvre labels used throughout the stratified analysis are computed from tracks interpolated to 1 Hz, and at low speed the derived heading is sensitive to the interpolation, so the boundary between the straight and mild-turn categories should be read as approximate. Finally, no inference-time measurement was made: the comparison establishes predictive accuracy but says nothing about whether any of these models can run within the latency budget of a shipboard system.

7.5 Future work

Three directions follow from the results above. The clearest is calibration: Section 7.1 shows that the gap between the oracle and committed predictions is large, so a better-ordered mode distribution would convert accuracy the model already possesses into accuracy a deployed system could use, without any change to the trajectory decoder. A second is to supply the regulatory structure that the present formulation leaves implicit — a COLREG-aware attention mask, or an encounter-type embedding, would let the model condition on the obligations that govern a manoeuvre rather than inferring them from kinematics alone. A third is joint rather than marginal prediction: the current formulation predicts each vessel independently while conditioning on its neighbours, which

cannot express the mutual consistency of two ships resolving an encounter between them. Underpinning all three is the need for a public maritime benchmark; the comparison reported here is reproducible in method but not in data, and the field will not converge on reliable baselines until an openly available AIS corpus with fixed splits and agreed metrics exists.

8 Conclusion

Published results in AIS trajectory prediction cannot be compared with one another: each study reports on a private regional dataset, at its own prediction horizon, under its own preprocessing and metric conventions. This paper addressed that gap by placing five architectures — an ordinary least-squares baseline, TrAISformer, AIS-ACNet, GAT-LSTM and a maritime adaptation of Wayformer — under a single evaluation protocol: the same corpus, the same 5-minute horizon at 1 Hz, the same splits, the same metric implementations, and a held-out test set of 11,891 prediction windows. The preprocessing pipeline, model configurations and evaluation scripts are released so that the comparison can be repeated and extended.

The comparison yields a result that the maritime literature does not currently surface. Three of the four learned architectures produce larger terminal errors than constant-velocity extrapolation of the observed track; only Wayformer-AIS improves on it, reaching 15.57 m minADE_6 and 31.71 m minFDE_6 against 26.70 m and 51.43 m for the baseline, with its margin widening as manoeuvre difficulty rises. Wayformer-AIS is also the only model in the comparison given the surrounding traffic as input rather than a single vessel's track, and among the designs examined it is the only one whose predictions are worth preferring to a straight line. Two qualifications travel with that result. The reported metrics select the best of six predicted modes after the fact, and the error of the trajectory the model actually commits to is roughly 40 % higher — a limitation of ranking rather than of representation, since the closer trajectory is generated but not ranked first. And no model examined here solves the task: a 10 m terminal error is exceeded in 67.3 % of test windows even under the favourable metric. Better calibration of the mode distribution, explicit encoding of COLREG structure, joint rather than marginal prediction, and above all an openly available AIS benchmark are the directions most likely to move those numbers.

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